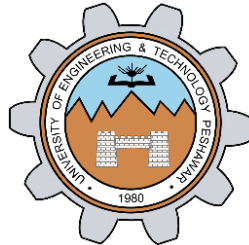


**LAB # 03**  
**INTERFACING PUSH BUTTONS TO 8051 DEVELOPMENT KIT**  
**USING POLLING**



**Spring 2024**  
**CSE-307L Microprocessor Based System Design Lab**

Submitted by: **Shahzad Bangash**

Registration No: **21PWCSE1980**

Class Section: **C**

“On my honor, as student at University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

**Engr. Shahzada Fahim Jan**

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Department of Computer Systems Engineering  
University of Engineering and Technology, Peshawar

## Objectives of The Lab:

- Understand the concept of polling.
- To learn the use of Sbit keyword.

## Polling:

The microcontroller continuously monitors status of a given device. When the condition meets, it performs the services. After that, it moves on to monitor the status of next device until everyone is serviced. It monitors several devices and serves each of them when certain conditions meet. It is not efficient method. This method wastes much of the microcontroller's cycle or time.

## Sbit (single bit):

It is a keyword used to access single-bit addressable registers.

## Lab Tasks:

### Task 1:

Turning Led on/off in 8051 microcontroller.

### C Code:

```
1  #include<reg51.h>
2  #include<stdio.h>
3  sbit Mybit=P1^0;
4  void main(void){
5      unsigned int i;
6      for(i=0; i<=500; i++)
7      {
8          Mybit=0;
9          Mybit=1;
10     }
11 }
12
13
```

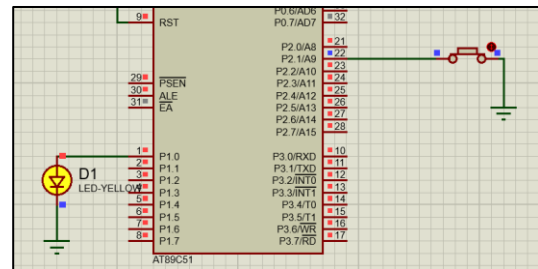
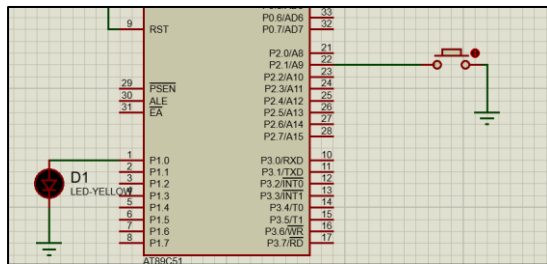
### Task 2:

Implementing Led on/off with push button.

### C Code:

```
1  #include <reg51.h>
2  #include <stdio.h>
3  sbit led1=P1^0;
4  sbit switch1 =P2^1;
5  void main(void)
6  {
7      while (1)
8      {
9
10         if(switch1==0)
11         {
12             led1=1;
13
14         }
15         else
16         {
17             led1=0;
18
19         }
20     }
21 }
22
23
24
```

## Simulation:



## Task 3:

Implementing 4 leds on/off with 2 push buttons.

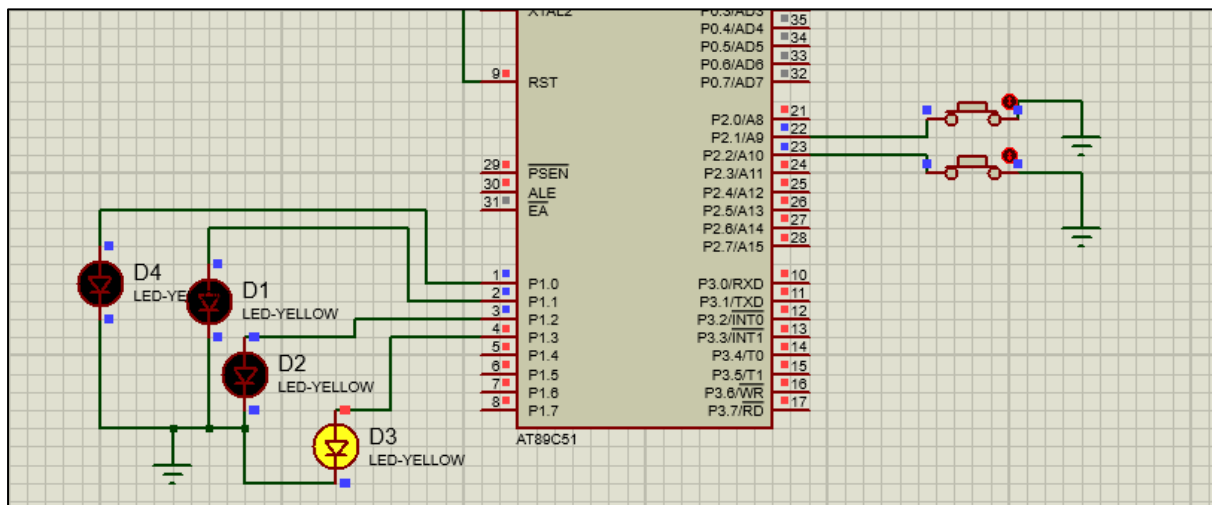
## C Code:

```

1  #include <reg51.h>
2  #include <stdio.h>
3  sbit led1=P1^0;
4  sbit led2=P1^1;
5  sbit led3=P1^2;
6  sbit led4=P1^3;
7  sbit switch1 =P2^1;
8  sbit switch2=P2^2;
9  void main(void)
10 {
11
12     while (1)
13     {
14
15         if (switch1==1 & switch2==1)
16         {
17             // P1=0x00;
18             led1=1; //p0.0
19             led2=0;
20             led3=0;
21             led4=0;
22
23         }
24         // P1=0x01;
25         else if (switch1==1 & switch2==0)
26         {
27             // P1=0x01;
28             led1=0;
29
30         }
31
32         // P1=0x01;
33         led1=0;
34         led2=1; //p0.1
35         led3=0;
36         led4=0;
37
38     }
39
40     else if (switch1==0 & switch2==1)
41     {
42         led1=0;
43         led2=0;
44         led3=1; //p0.2
45         led4=0;
46
47     }
48
49     else
50     {
51         // P1=0x01;
52         led1=0;
53         led2=0;
54         led3=0;
55         led4=1; //p0.3
56     }
57 }

```

## Simulation:



## Task 4:

Attach 3 push button with PORT2 and 8 LED's with PORT1 and turn ON led when the corresponding buttons are pressed, use polling for monitoring the buttons.

## C Code:

```
1 #include <reg51.h>
2 sbit led1=P1^0;
3 sbit led2=P1^1;
4 sbit led3=P1^2;
5 sbit led4=P1^3;
6 sbit led5=P1^4;
7 sbit led6=P1^5;
8 sbit led7=P1^6;
9 sbit led8=P1^7;
10 sbit pbtn1=P2^0;
11 sbit pbtn2=P2^1;
12 sbit pbtn3=P2^2;
13 void main(void){
14     while (1){
15         if(pbtn3==0 & pbtn2==0 & pbtn1==1){
16             led1=0;
17             led2=1;
18             led3=0;
19             led4=0;
20             led5=0;
21             led6=0;
22             led7=0;
23             led8=0;
24     }
```

```
25 else if(pbtn3==0 & pbtn2==1 & pbtn1==0){
26     led1=0;
27     led2=0;
28     led3=1;
29     led4=0;
30     led5=0;
31     led6=0;
32     led7=0;
33     led8=0;
34 }
35 else if(pbtn3==0 & pbtn2==1 & pbtn1==1){
36     led1=0;
37     led2=0;
38     led3=0;
39     led4=1;
40     led5=0;
41     led6=0;
42     led7=0;
43     led8=0;
44 }
```

```
45 else if(pbtn3==1 & pbtn2==0 & pbtn1==0){
46     led1=0;
47     led2=0;
48     led3=0;
49     led4=0;
50     led5=1;
51     led6=0;
52     led7=0;
53     led8=0;
54 }
55 else if(pbtn3==1 & pbtn2==0 & pbtn1==1){
56     led1=0;
57     led2=0;
58     led3=0;
59     led4=0;
60     led5=0;
61     led6=1;
62     led7=0;
63     led8=0;
64 }
```

```
65 else if(pbtn3==1 & pbtn2==1 & pbtn1==0){
66     led1=0;
67     led2=0;
68     led3=0;
69     led4=0;
70     led5=0;
71     led6=0;
72     led7=1;
73     led8=0;
74 }
75 else if(pbtn3==1 & pbtn2==1 & pbtn1==1){
76     led1=0;
77     led2=0;
78     led3=0;
79     led4=0;
80     led5=0;
81     led6=0;
82     led7=0;
83     led8=1;
84 }else{
```

```

84     }else{
85         led1=1;
86         led2=0;
87         led3=0;
88         led4=0;
89         led5=0;
90         led6=0;
91         led7=0;
92         led8=0;
93     }
94 }
95 }

```

## Simulation in Proteus:

The following pictures show the simulation of 8 leds in proteus with different combinations.

